Advanced Calculus I Hand-in Assignment 5

Student ID Number (學號): Name :

- This Assignment **will be collected in class.** (這份作業會在上課的時候收。) Please put your assignment on the desk in our classroom **before 10:25 on Oct 21th, Wednesday**. (請在10月21日10:25分以前，將作業放在教室的講台。) Please write down your student ID number, name and answer on A4 papers and staple your papers together （請將學號、姓名、答案寫在A4大小的紙張。若有兩張以上的答案紙，請用訂書機訂好。).
- Note that we do not collect late assignment. (請注意，不收遲交作業。)
- 作業在第2頁附上中文，希望可以在學習英文上對同學有所幫助，但是小考、期中、期末考不會附上中文，**建議同學先讀第一頁全英文的部份，再比較理解的情況和第二頁有什麼不同**。寫作業、考試可以寫中文或是中文、英文混合。

Problem 1. (10 points.) Suppose that $\{x_k\}$ is a sequence of points in $\mathbb{R}^n$, $\alpha \in \mathbb{R}$ and $\{x_k\}$ is convergent. Please show that

$$\lim_{k\to\infty} \alpha x_k = \alpha \lim_{k\to\infty} x_k$$

Problem 2. (10 points) Let $x_k \in \mathbb{R}^n$. Then $x_k \to L$, as $k \to \infty$ if and only if for every open set V which contains L there is an $N \in \mathbb{N}$ such that $k \geq N$ implies $x_k \in V$.

Bonus Problem. (3 points) Let $\{x_k\}$ be a sequence of points in $\mathbb{R}^n$ Please write down the definition of the following terminology.
(1) $\{x_k\}$ *converges* to some point $a \in \mathbb{R}^n$.
(2) $\{x_k\}$ is *bounded*.
(3) A *subsequence* of $\{x_k\}$.

Problem 1. (10 分.)假設$\{x_k\}$爲$\mathbb{R}^n$空間中的數列，$\alpha \in \mathbb{R}$且$\{x_k\}$收斂。請證明

$$\lim_{k\to\infty} \alpha x_k = \alpha \lim_{k\to\infty} x_k$$

Problem 2. (10 分) 令$x_k \in \mathbb{R}^n$。則$x_k \to L$, 當$k \to \infty$ 若且爲若對於所有包含L的開集V，會存在$N \in \mathbb{N}$使得$k \geq N$會導致$x_k \in V$。

Bonus Problem. (3 points) 令$\{x_k\}$爲$\mathbb{R}^n$空間的數列。請寫出下列術語的定義。
(1) $\{x_k\}$ 收斂到某個點$a \in \mathbb{R}^n$。
(2) $\{x_k\}$ 爲有界限。
(3) 一個$\{x_k\}$的子數列。